Queue Implementation Using Linked List

C Program Code and Execution Output

PROGRAM SOURCE CODE

```
#include<stdio.h>
#include<stdlib.h>

/* Structure to create a node with data and next pointer */
struct node
{
    int data;
    struct node *next;
};

struct node *front = NULL;
struct node *rear = NULL;

/* Enqueue operation on a queue */
void enqueue(int value)
{
    struct node *ptr;
    ptr = (struct node*)malloc(sizeof(struct node));
    ptr->data = value;
    ptr->next = NULL;
    if((front == NULL) && (rear == NULL))
    {
        front = rear = ptr;
    }
    else
    {
        rear->next = ptr;
        rear = ptr;
    }
    printf("Node is Inserted:%d\n\n", value);
}

/* Dequeue operation on a queue */
int dequeue()
{
    if(front == NULL)
    {
        printf("\nUnderflow\n");
        return -1;
    }
    else
    {
        struct node *temp = front;
        int temp_data = front->data;
        front = front->next;
        free(temp);
        return temp_data;
    }
}

/* Display all elements of queue */
void display()
{
```

PROGRAM OUTPUT

```
Node is Inserted:10

Node is Inserted:20

Node is Inserted:30

The queue is
10--->20--->30--->NULL

The queue is
20--->30--->NULL
```